IBT × GGATEWAY DATA SCIENCE BOOTCAMP

Project Documentation Report

Early Prediction of Clinical Deterioration in ICU Patients

Using Machine Learning

Supervised Learning · Binary Classification · Healthcare AI

Mentor	Track	Type	Year
Courage Dike	**Supervised Learning**	**Binary Classification**	**2026**

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01 Executive Summary

This document presents the complete technical documentation for the ICU Clinical Deterioration Prediction project, developed as the capstone for the IBT × GGateway Data Science and Machine Learning Bootcamp under the supervision of Mentor Courage Dike.

The project addresses a critical clinical challenge: patients admitted to Intensive Care Units (ICUs) frequently experience sudden deterioration requiring mechanical ventilation. Current monitoring systems display real-time vital signs but cannot predict deterioration hours in advance, limiting the window for clinical intervention.

Our team built a binary machine learning classifier trained on the eICU Collaborative Research Database (PhysioNet). The model predicts whether an ICU patient is at high risk of requiring mechanical ventilation within the next 6–12 hours, using rolling windows of vital signs, respiratory parameters, treatment escalation flags, and patient demographics.

Key Project Metrics

Metric	Value
Dataset	eICU Collaborative Research Database (PhysioNet)
ICU Patients	2,520 unique ICU stays
Total Hourly Windows	136,867 rows
Training Set	109,474 rows — 250 positives (0.23%)
Test Set	27,393 rows — 48 positives (0.18%)
Class Imbalance Ratio	436:1 (severe imbalance)
Features Used	31 engineered features
Best Model	Random Forest + class_weight='balanced_subsample'
Best AUROC	0.645
Best Recall	50% of true ventilation windows detected
Primary Metric	AUPRC (Area Under Precision-Recall Curve)

02 Background

Patients admitted to the ICU are among the most critically ill in any hospital setting. They are continuously monitored for physiological changes that may indicate worsening condition. Despite this monitoring, sudden clinical deterioration remains a leading cause of preventable ICU mortality.

Mechanical ventilation — the use of a machine to assist or replace spontaneous breathing — is one of the most common life-saving interventions in the ICU. It is typically initiated when a patient develops respiratory failure, severe hemodynamic instability, or loss of airway protection. However, the physiological decline that necessitates ventilation often begins hours before the intervention becomes unavoidable.

Early Warning Scores (EWS) such as NEWS2 and SOFA have been developed to identify at-risk patients. These tools rely on static, snapshot-based thresholds evaluated manually at fixed intervals. They do not capture temporal trends, do not continuously update, and cannot leverage the complexity of available electronic health record data.

> **Research Question: Can we predict whether an ICU patient is at high risk of requiring mechanical ventilation within the next 6–12 hours using historical vital signs and clinical data from the eICU dataset?**

Why This Matters

- Each year, preventable ICU deaths are associated with delayed detection of respiratory failure
- Mechanical ventilation, when initiated proactively, leads to better outcomes than emergency intubation
- A 6–12 hour advance warning gives clinical teams time to prepare, escalate care, and consult specialists
- Automated continuous prediction removes dependence on manual nursing-administered scoring

03 Problem Statement

3.1 Clinical Problem

Current ICU monitoring systems generate continuous streams of physiological data but present it as real-time snapshots. Clinicians must manually interpret trends across multiple parameters simultaneously, often under significant time pressure and cognitive load. This reactive approach means interventions are frequently initiated after deterioration is already well underway.

3.2 Technical Formulation

This project formulates the problem as a supervised binary classification task on an hourly sliding window basis:

Class	Label	Definition
Stable	y = 0	The patient does not require mechanical ventilation in the next 6–12 hours
High Risk	y = 1	Mechanical ventilation will be initiated within the next 6–12 hours

3.3 Why the 6–12 Hour Window?

- Beyond 12 hours: The physiological signal is too weak — vital signs have not yet begun their characteristic deterioration pattern
- Within 6 hours: Too late for meaningful clinical preparation; also introduces data leakage risk from ventilator-related measurements already in use
- 6–12 hours: Provides actionable lead time for clinical teams to prepare, escalate care, and potentially prevent emergency intubation

3.4 Key Challenges

Challenge	Description	Solution
Extreme Class Imbalance	Only 0.22% of hourly windows are positive (436:1 ratio)	class_weight, scale_pos_weight, SMOTE variants
Data Leakage	Ventilator settings recorded after intubation contaminate features	Strict time-window filtering with merge_asof
Temporal Structure	Patient data is time-series — row-level split leaks patient info	Patient-level train/test split
Sparse Signal	250 true positive windows out of 136,867 total	AUPRC as primary metric; tree-based models

04 Objectives

4.1 Primary Objective

Develop a machine learning pipeline that continuously evaluates ICU patients at hourly intervals and predicts the probability of clinical deterioration — specifically the need for mechanical ventilation — within the next 6–12 hours.

4.2 Specific Technical Objectives

1. Build an hourly patient evaluation grid from raw eICU time-series data, converting each ICU stay into a sequence of hourly observation windows.
2. Engineer clinically meaningful features from rolling 3-hour windows of vital signs, respiratory parameters, and treatment records, capturing both absolute values and temporal trends.
3. Create binary labels (y_high) with strict leakage prevention, ensuring no post-ventilation data contaminates the training signal.
4. Train and evaluate Logistic Regression, Random Forest, and XGBoost models using AUPRC as the primary metric.
5. Compare all model-strategy combinations and identify the best configuration for clinical deployment.
6. Interpret model decisions using feature importance to ensure clinical plausibility of predictions.

4.3 Success Criteria

Primary Metric	AUPRC meaningfully above random baseline (0.0022) — indicates the model captures real deterioration signal
Secondary Metric	AUROC > 0.60 — model discriminates between safe and danger hours better than chance
Clinical Recall Goal	Recall ≥ 30% — missing danger hours is clinically more costly than false alarms
Interpretability	Feature importance rankings should align with established clinical knowledge
Reproducibility	All notebooks run top-to-bottom without errors; random seeds fixed at 42

05 Dataset Description

5.1 Data Source

The project uses the eICU Collaborative Research Database (eICU-CRD), a multi-center critical care database developed by Philips Healthcare in partnership with MIT Laboratory for Computational Physiology. The database is hosted on PhysioNet and requires CITI training and a signed Data Use Agreement for access.

Database Name	eICU Collaborative Research Database (eICU-CRD)
Host Platform	PhysioNet — physionet.org
Access Level	Credentialed — CITI training + Data Use Agreement required
ICU Stays Used	2,520 unique patient ICU stays
Ventilated Patients	69 (2.7% ventilation rate)
Data Collection Period	2014–2015
Geographic Coverage	Multiple US hospitals and health systems
Primary Reference	Pollard et al. (2018), Scientific Data

5.2 Source Tables

Table	Rows	Primary Use in Project
patient.csv	2,520	Patient demographics, ICU stay duration, discharge status, unit type
respiratoryCare.csv	5,436	Ventilation start times — used exclusively for label creation
respiratoryCharting.csv	169,817	FiO2, PEEP, Tidal Volume, RR — key respiratory indicators
vitalPeriodic.csv	1,633,310	Periodic vitals: heart rate, SpO2 (sao2), respiration rate
treatment.csv	38,048	Treatment flags: oxygen therapy, noninvasive ventilation

5.3 Key Variables

Variable	Source	Type	Clinical Role
heartrate	vitalPeriodic	Continuous	Core vital — tachycardia precedes deterioration
sao2 (SpO2)	vitalPeriodic	Continuous	Oxygen saturation — drops before respiratory failure
respiration	vitalPeriodic	Continuous	Respiratory rate — rises with distress
FiO2	respiratoryCharting	Continuous	Best single predictor (corr = 0.016 with y_high)

PEEP	respiratoryCharting	Continuous	Key ventilator parameter — 86% coverage of vent patients
ventstartoffset	respiratoryCare	Integer (min)	Label source only — NOT used as a feature
on_oxygen_therapy	treatment	Binary flag	74% of ventilated patients received oxygen
on_noninvasive_vent	treatment	Binary flag	Pre-intubation escalation signal
age, gender	patient	Mixed	Baseline demographics
unittype, admissionweight	patient	Mixed	Clinical context and baseline risk

06 Exploratory Data Analysis

6.1 Overview

A comprehensive EDA was conducted across all five source tables before any modeling. The analysis covered data quality (missing values, outliers, duplicates), target variable characteristics, univariate distributions, bivariate comparisons between ventilated and non-ventilated patients, and multivariate correlations with the target.

6.2 Missing Data Analysis

Column	Missing %	Decision & Rationale
patient — dischargeweight	51.0%	Dropped — post-ICU measurement, not predictive of deterioration
patient — hospitaladmitsource	23.6%	Mode imputation — categorical baseline feature
patient — apacheadmissiondx	11.9%	Mode imputation — admission diagnosis category
patient — admissionweight	7.9%	Median imputation — clinical baseline retained
vital_periodic — icp, pasystolic, padiastolic	98–99%	Dropped — near-entirely missing, no usable signal
vital_periodic — temperature, cvp, etco2	87–96%	Dropped — too sparse for reliable modeling
vital_periodic — heartrate	0.4%	Retained — highly complete, imputed with median
vital_periodic — sao2	11.8%	Retained with median imputation
vital_periodic — respiration	15.5%	Retained with median imputation
resp_care — 20+ ventilator columns	90–100%	All dropped — entirely missing or near-zero fill

6.3 Target Variable Analysis

Metric	Value	Implication
Total ICU patients	2,520	Working dataset size
Ventilated patients	69 (2.7%)	Very rare event — imbalance challenge
Non-ventilated patients	2,451 (97.3%)	Dominant class
Median ventilation onset	~27 hours after admission	Early-stay event
Hourly windows (y=1)	298 out of 136,867 (0.22%)	Extreme imbalance in the ML dataset
Imbalance ratio	~436:1	Justifies AUPRC as primary metric

6.4 Univariate Analysis — Key Findings

Feature	Finding	Decision
Age	Peaks at 65–70 years — typical ICU population; well distributed	Keep
Admission Height	Max 600cm — physiologically impossible; cleaned to 100–250cm range	Keep after cleaning
Admission Weight	Max 515kg — impossible; cleaned to 20–300kg range	Keep after cleaning
Heart Rate	Normal distribution ~80bpm; 335 readings ≤0 removed	Keep, clean extremes
SpO2 (sao2)	Most values 95–100%; 1,985 readings <50% removed as impossible	Keep, clean <50%
Respiration Rate	Normal 15–25 br/min; outliers >80 and ≤0 removed	Keep, clean extremes
FiO2	27,393 records; 96% coverage of ventilated patients	High-priority feature
PEEP	13,363 records; 86% coverage of ventilated patients	High-priority feature
ICU Unit Type	CSICU (4.6%), CTICU (3.8%), Cardiac ICU (3.8%) highest vent rates	Keep for context

6.5 Bivariate Analysis — Safe vs. Danger Hours

After building the hourly grid with engineered labels, a direct comparison of feature means between safe (y=0) and danger (y=1) windows confirmed that all key features move in the clinically expected direction:

Feature	Safe (y=0)	Danger (y=1)	Direction	Clinical Meaning
Heart Rate (bpm)	85.3	90.7	↑ +6.3%	Tachycardia developing before ventilation
SpO2 (%)	96.6	95.8	↓ −0.8%	Oxygen saturation dropping
Respiration Rate	20.0	22.3	↑ +11.5%	Respiratory distress pattern
FiO2 (%)	41.1	47.2	↑ +14.8%	Patient needs increasing supplemental oxygen
SpO2 Slope (3h)	−0.04	−0.21	↓ faster	SpO2 dropping faster in danger hours
HR Slope (3h)	−0.14	−0.58	↓ faster	HR trend more volatile before ventilation

> All vital signs move in the clinically correct direction during danger hours — validating that the engineered features capture real physiological deterioration patterns.

6.6 Feature Coverage for Ventilated Patients

Feature	Coverage	Decision
FiO2	96% — 66 of 69 patients	High priority — include
PEEP	86% — 59 of 69 patients	High priority — include
Tidal Volume (set)	84% — 57 of 69 patients	Include
On Oxygen Therapy	74% — 51 of 69 patients	Include as binary flag
On Noninvasive Vent	23% — 16 of 69 patients	Include — escalation signal
RR from resp_chart	35% — 24 of 69 patients	Skip — already have from vitalPeriodic
Bronchodilator	19% — 13 of 69 patients	Skip — too sparse

6.7 Data Leakage Check

Signal	Before Ventilation	After Ventilation	Strategy
FiO2	26% of readings	74% of readings	Use only pre-vent readings via merge_asof
PEEP	26% of readings	74% of readings	Use only pre-vent readings via merge_asof
Tidal Volume	27% of readings	73% of readings	Use only pre-vent readings via merge_asof

07 Data Preprocessing

7.1 Pipeline Overview

All preprocessing was performed with strict train/test separation. Every statistic — medians, modes, scaler parameters, encoder categories — was computed exclusively on training data and applied to test data without re-fitting, preventing any form of information leakage.

1 **Step 1 — Outlier Treatment**
Physiologically impossible values replaced: heart rate ≤0 or >250bpm; SpO2 <50%; respiration ≤0 or >80 br/min; admission height >250cm or <100cm; admission weight >300kg or <20kg. All replaced with training-set median after splitting.

2 **Step 2 — Missing Value Imputation**
Numeric columns imputed with training-set median via SimpleImputer. Categorical columns imputed with training-set mode. Strategy fitted on train only, applied to both train and test. Result: zero missing values across all five tables.

3 **Step 3 — Categorical Encoding**
ICU unit type (8 categories) and gender (2 categories) one-hot encoded using OneHotEncoder(handle_unknown='ignore'). Encoder fitted only on training data. Added 10 binary columns to feature matrix.

4 **Step 4 — Feature Scaling**
StandardScaler applied to 16 continuous numeric features. Scaler fitted on train, transformation applied to both. Binary and flag columns left unscaled. Required for Logistic Regression; applied consistently to all models for fair comparison.

7.2 Class Imbalance Strategies

Strategy	How It Works	Applied To	Key Trade-off
class_weight='balanced'	Re-weights loss function — penalizes misclassifying positives more	Logistic Regression, Random Forest	Uses real data only; no synthetic samples
scale_pos_weight=436.9	XGBoost native equivalent of class_weight	XGBoost only	Handles imbalance natively during boosting
SMOTE	Generates synthetic minority examples by interpolating between real ones	RF and XGBoost variants	Synthetic data may not reflect real clinical trajectories

SMOTE+Tomek	SMOTE oversampling + removal of ambiguous majority samples	RF and XGBoost variants	No improvement observed in this study
ADASYN	Adaptive SMOTE — focuses on harder-to-classify examples	RF and XGBoost variants	Best SMOTE variant in this study

08 Feature Engineering

8.1 Hourly Patient Grid

The core innovation of this project's data pipeline is the conversion of each patient's ICU stay into an hourly evaluation grid. Rather than generating one prediction per patient, the model evaluates each patient at every hour of their stay — creating a time-aware training set.

Grid Construction	Each patient's stay divided into 1-hour windows from admission to discharge
Time Resolution	60 minutes per window
Total Windows	136,867 rows across 2,494 patients after leakage removal
Alignment Method	merge_asof with direction='backward' — carries forward last known value to each window
Leakage Removal	All windows from 6 hours before ventilation onward removed from the dataset
Leakage Column	vent_start_min dropped from features (directly related to y_high by construction)

8.2 Label Definition (y_high)

- y_high = 1: Window falls in [vent_start_min − 720, vent_start_min − 360] — the 6–12h danger zone
- y_high = 0: All other windows — non-ventilated patients and windows more than 12h before ventilation
- Excluded: Windows from vent_start_min − 360 onward — too close, leakage zone
- Result: 298 positive windows out of 136,867 total (0.22% positive rate)

8.3 Rolling Window Features

Feature Group	Features Created	Clinical Rationale
Last known value	heartrate_last, sao2_last, respiration_last	Most recent reading available at decision time
3h Rolling Mean	heartrate_mean_3h, sao2_mean_3h, respiration_mean_3h	Smoothed recent trend — reduces noise from single readings
3h Rolling Std	heartrate_std_3h, sao2_std_3h, respiration_std_3h	Variability — high variance signals physiological instability
3h Slope (diff)	heartrate_slope_3h, sao2_slope_3h, respiration_slope_3h	Rate of change — captures velocity of deterioration

Respiratory params	fio2_last, peep_last	Oxygen demand and pre-intubation ventilator parameters
Treatment flags	on_oxygen_therapy, on_noninvasive_vent	Escalating care — signals of increasing respiratory support
Missingness indicators	hr_missing, sao2_missing, fio2_missing	Missing vitals in ICU are clinically informative
Demographics	age, admissionweight, unittype (8 OHE), gender (2 OHE)	Baseline risk context and ICU unit type

8.4 Final Feature Set Summary

Category	Count	Examples
Vital sign last values	3	heartrate_last, sao2_last, respiration_last
3h Rolling mean	3	heartrate_mean_3h, sao2_mean_3h, respiration_mean_3h
3h Rolling std	3	heartrate_std_3h, sao2_std_3h, respiration_std_3h
3h Slope	3	heartrate_slope_3h, sao2_slope_3h, respiration_slope_3h
Respiratory parameters	2	fio2_last, peep_last
Treatment flags	2	on_oxygen_therapy, on_noninvasive_vent
Missingness indicators	3	hr_missing, sao2_missing, fio2_missing
Demographics (numeric)	2	age, admissionweight
Unit type (one-hot)	8	unittype_MICU, unittype_SICU, unittype_CCU-CTICU, etc.
Gender (one-hot)	2	gender_Female, gender_Male
TOTAL	31	Complete feature matrix for model training

09 Model Development

9.1 Train / Test Split

A patient-level split was used rather than a row-level split. This is a critical methodological choice: splitting by row allows the same patient's hours to appear in both training and test sets, causing the model to memorize patient-specific patterns rather than learning generalizable features — a form of data leakage.

Split Method	train_test_split on unique patientunitstayid values
Test Proportion	20% of patients held out
Training Set	1,995 patients → 109,474 hourly windows → 250 positive labels (0.23%)
Test Set	499 patients → 27,393 hourly windows → 48 positive labels (0.18%)
Random Seed	42 (fully reproducible)

9.2 Model 1 — Logistic Regression (Baseline)

Logistic Regression serves as the required baseline. It finds a linear decision boundary between safe and danger hours. Its primary value is establishing a performance floor and confirming that the preprocessing pipeline is correct.

Algorithm	LogisticRegression (scikit-learn)
Imbalance Strategy	class_weight='balanced'
Solver	lbfgs	max_iter: 1000
Key Limitation	Assumes linear separability — misses interaction effects between features
Result	AUPRC: 0.0015	AUROC: 0.400	Recall: 71%	F1: 0.00

Logistic Regression confirms that the preprocessing pipeline is technically correct, but reveals the fundamental limitation of a linear model on this problem. The high recall (71%) is misleading — it is achieved by lowering the decision threshold so aggressively that the model predicts positive for a large fraction of all windows, generating excessive false alarms. The near-zero F1 and AUROC below 0.5 indicate the model is unable to find a meaningful linear boundary between safe and danger hours in the 31-feature space. The model correctly identifies who is at risk (high recall) but cannot discriminate when they are at risk, confirming that non-linear models are required for this task.

9.3 Model 2 — Random Forest

Random Forest builds an ensemble of decision trees, each trained on a random bootstrap sample of data and random subset of features. Predictions are aggregated by probability averaging. It handles non-linear relationships, mixed feature types, and class imbalance naturally.

Algorithm	RandomForestClassifier (scikit-learn)
Strategies Tested	class_weight='balanced_subsample' | SMOTE | SMOTE+Tomek | ADASYN
Number of Trees	100 (default)
Best Configuration	class_weight='balanced_subsample'
Best Result	AUPRC: 0.0031 | AUROC: 0.645 | Recall: 50% | F1: 0.011

Random Forest is the best-performing model in this study. Its ensemble of 300 trees, each trained on a bootstrap sample with class_weight='balanced_subsample', correctly identifies 50% of all true ventilation risk windows — 6 to 12 hours in advance — without generating any synthetic training data. The AUROC of 0.645 demonstrates meaningful discrimination well above the 0.5 random baseline. Importantly, class_weight='balanced_subsample' outperforms SMOTE variants for this model: while SMOTE variants raise AUPRC slightly to 0.0040, they significantly reduce recall from 50% to 31%, trading away the most clinically critical metric. The top feature importances — age, admissionweight, heartrate_mean_3h, fio2_last — are clinically consistent with known ICU deterioration patterns.

9.4 Model 3 — XGBoost

XGBoost builds trees sequentially — each new tree corrects the prediction errors of the current ensemble (gradient boosting). It is the gold standard algorithm for tabular clinical data in published ICU prediction research.

Algorithm	XGBClassifier (xgboost library)
Strategies Tested	scale_pos_weight=436.9 | SMOTE | SMOTE+Tomek | ADASYN
eval_metric	aucpr — optimizes directly for AUPRC during tree construction
Best Configuration	scale_pos_weight=436.9 (native imbalance handling)
Best Result	AUPRC: 0.0032 | AUROC: 0.632 | Recall: 17% | F1: 0.012
Key Finding	SMOTE hurt XGBoost — recall dropped from 17% to 4% with basic SMOTE

XGBoost demonstrated that native class weighting (scale_pos_weight) outperforms synthetic oversampling on this dataset. With only 250 positive training rows, SMOTE generates noisy interpolations that dilute the real deterioration signal, causing recall to collapse from 17% to 4%. ADASYN partially recovers by focusing synthetic examples on hard-to-classify boundary cases, achieving AUROC 0.636 — still below the unaugmented baseline. XGBoost's early stopping triggered at round 18 out of 500, suggesting the model quickly exhausts the learnable signal available in 3-hour rolling features.

9.5 Model 4 — LightGBM

LightGBM (Light Gradient Boosting Machine) is a highly optimised gradient boosting framework developed by Microsoft. It uses histogram-based leaf-wise tree growth, making it significantly faster than XGBoost while achieving comparable or superior accuracy on tabular clinical data. Unlike XGBoost's level-wise growth, LightGBM's leaf-wise strategy reduces loss more aggressively per tree, making it well-suited for learning from the rare positive class. It was confirmed as a top performer in the ICD-Balancer benchmark, which used the eICU dataset.

Algorithm	LGBMClassifier (lightgbm library)

Imbalance Strategy	scale_pos_weight=436.9 | Effective Number of Samples weighting (beta=0.9999) | SMOTE
Tuning Method	Optuna Bayesian optimisation (50 trials, 3-fold stratified CV) | early_stopping (50 rounds)
Best Configuration	Optuna-tuned + Effective Sample Weighting + scale_pos_weight=436.9
Baseline Result	AUPRC: 0.0037 | AUROC: 0.637 | 2.1× above random baseline
Key Finding	Effective Number of Samples weighting outperformed standard class_weight; static features (age, weight) dominated importance, masking real-time vital signs

LightGBM introduced two important innovations over the previous models. First, Effective Number of Samples weighting (Cui et al., 2019) was applied — a mathematically smoother alternative to inverse frequency weighting that avoids over-penalising extremely rare classes. Second, Optuna Bayesian hyperparameter tuning (50 trials, 3-fold CV) was used to systematically search the parameter space, which is more efficient than manual tuning. The baseline result (AUPRC 0.0037) was 2.1× above random, the highest absolute AUPRC of any single non-SMOTE model. Feature importance analysis revealed that static demographics (age, admissionweight) dominated the top features — a signal that the model was profiling patient risk level rather than detecting hour-by-hour deterioration, motivating the removal of these features in future iterations.

9.6 Deep Learning — Stacked LSTM & Bidirectional LSTM

As a deep learning extension, two recurrent neural network architectures were implemented using TensorFlow/Keras and tuned with keras_tuner RandomSearch. Both models were trained on the same 31-feature hourly grid as the tree-based models, with seq_len=1. A Stacked LSTM places multiple LSTM layers on top of each other, learning progressively abstract temporal representations. A Bidirectional LSTM (BiLSTM) processes sequences in both forward and backward directions, concatenating outputs to give the model access to broader context at every timestep.

Property	Stacked LSTM	Bidirectional LSTM
Architecture	LSTM(128) → Dropout(0.1) → LSTM(32) → Dropout(0.4) → Dense(1, sigmoid)	BiLSTM(128×2=256) → Dropout(0.1) → BiLSTM(32×2=64) → Dropout(0.4) → Dense(1, sigmoid)
Tuning Method	keras_tuner RandomSearch — 10 trials, EarlyStopping(patience=3), 30 epochs	keras_tuner RandomSearch — 10 trials, EarlyStopping(patience=3), 30 epochs
AUROC	0.5027 (near random)	0.5377 (slightly above random)
AUPRC	0.0030 (above random baseline 0.0022)	0.0026 (slightly below Stacked LSTM)
Key Limitation	seq_len=1 means no real temporal sequences are fed to the LSTM layers — both models learn from tabular snapshots, not true time-series. Confusion matrices show all predictions classified as negative.	

Both LSTM models confirmed the hypothesis that deep learning architectures do not outperform tree-based models on tabular ICU data with seq_len=1. Without true sequential input (multiple consecutive hourly readings fed as a sequence), LSTM cells cannot exploit their temporal memory mechanism — they effectively reduce to dense layers. The BiLSTM achieved marginally higher AUROC (0.5377 vs 0.5027) due to its wider architecture, but lower AUPRC (0.0026 vs 0.0030), indicating it is less precise on the minority class. Both models predicted no positive cases at the default threshold,

confirming that deep learning on this dataset requires either longer input sequences (seq_len ≥ 12) or significantly more training data before it can compete with Random Forest or LightGBM.

10 Evaluation & Results

10.1 Evaluation Metrics

A model that predicts y=0 (stable) for every hourly window achieves 99.78% accuracy — yet identifies zero patients at risk. AUPRC (Area Under Precision-Recall Curve) is the correct metric for severely imbalanced clinical datasets.

AUPRC (Primary)	Area Under Precision-Recall Curve — focuses on the positive (danger) class
AUROC (Secondary)	Area Under ROC Curve — overall discrimination; used for literature comparison
Recall	Proportion of true ventilation windows correctly flagged — clinically most important
F1 Score	Harmonic mean of precision and recall
Random Baseline AUPRC	0.0022 (at 0.22% positive rate — any meaningful model must exceed this)

10.2 Full Model Comparison

Model	Strategy	AUPRC	AUROC	Recall	F1	Rank
Logistic Regression	class_weight=balanced	0.0015	0.400	71%	0.00	—
XGBoost	scale_pos_weight=436.9	0.0032	0.632	17%	0.012	
XGBoost	SMOTE	0.0030	0.590	4%	0.020	
XGBoost	SMOTE+Tomek	0.0030	0.590	4%	0.020	
XGBoost	ADASYN	0.0030	0.590	27%	0.020	
Random Forest	class_weight=balanced_sub.	0.0031	0.645	50%	0.011	Best Overall
Random Forest	SMOTE	0.0040	0.663	31%	0.014	
Random Forest	SMOTE+Tomek	0.0040	0.663	31%	0.014	
Random Forest	ADASYN	0.0040	0.663	31%	0.014	
LightGBM	scale_pos_weight=436.9 + Eff. Sampling	0.0037	0.637	23%	0.014	Highest AUPRC (non-SMOTE)
Stacked LSTM	RandomSearch Tuning, seq_len=1	0.0030	0.503	0%	0.00	—
Bidirectional LSTM	RandomSearch Tuning, seq_len=1	0.0026	0.538	0%	0.00	—

Across all 12 model-strategy combinations, Random Forest with class_weight='balanced_subsample' remains the best overall model by clinical criteria (AUROC 0.645, Recall 50%). LightGBM achieves the highest raw AUPRC (0.0037) among

non-SMOTE configurations, confirming its strength on tabular imbalanced data, but at lower recall than Random Forest. Both LSTM architectures underperform tree-based models under the current tabular, seq_len=1 setup — their AUROC values near 0.5 and zero recall confirm they require true sequential multi-timestep input to leverage their temporal memory. XGBoost with ADASYN is the strongest SMOTE-augmented single model (AUPRC 0.0043), but SMOTE variants consistently trade recall for marginal AUPRC gains, which is clinically unfavourable for a missed-ventilation-is-costly task.

Best Overall Model — Random Forest + class_weight='balanced_subsample'

- AUPRC: 0.0031 | AUROC: 0.645 | Recall: 50% | F1: 0.011
- Correctly identifies 50% of all true ventilation risk windows — 6–12 hours in advance
- AUROC 0.645 demonstrates meaningful discrimination above the 0.5 random baseline
- Uses only real patient data — no synthetic oversampling required

10.3 Feature Importance

Rank	Feature	Correlation with y_high	Clinical Interpretation
1	respiration_mean_3h	+0.0202	Mean RR over last 3h — respiratory distress building
2	respiration_last	+0.0179	Most recent respiratory rate reading
3	fio2_last	+0.0159	Last FiO2 value — oxygen demand indicator
4	heartrate_mean_3h	+0.0145	Mean HR over last 3h — tachycardia developing
5	sao2_missing	+0.0144	SpO2 missingness — informative in ICU
6	hr_missing	+0.0144	HR missingness — informative in ICU
7	heartrate_last	+0.0134	Most recent heart rate reading
8	sao2_mean_3h	−0.0126	Mean SpO2 (lower = more danger)
9	sao2_last	−0.0115	Most recent SpO2 (lower = more danger)
10	admissionweight	+0.0053	Body mass affects respiratory mechanics

10.4 Key Analytical Findings

- SMOTE hurt XGBoost: Recall dropped from 17% to 4% with basic SMOTE. XGBoost handles imbalance better via scale_pos_weight internally — SMOTE creates noisy synthetic samples with only 250 positives in the training set.
- SMOTE improved Random Forest AUPRC slightly (0.0040 vs 0.0031) but significantly reduced recall from 50% to 31% — trading off the most clinically critical metric.
- ADASYN was consistently the best-performing SMOTE variant across both models, outperforming basic SMOTE and SMOTE+Tomek.
- SMOTE+Tomek produced identical results to SMOTE in this study — suggesting no meaningful boundary-ambiguous samples exist between classes.
- Logistic Regression achieved high recall (71%) but near-zero F1 — it predicts positive too liberally, generating excessive false alarms that render it clinically impractical.
- Precision ceiling at ~1%: A hard ceiling imposed by available features, not model architecture. Extending feature windows to 6h and 12h is the most impactful next step.

10.5 Why Models Hit a Precision Ceiling

Root Cause	Evidence	Recommended Fix
Feature window too short	3h rolling features miss slower deterioration trajectories	Extend windows to 6h and 12h
Demographic dominance	age and admissionweight rank in top features over vital sign dynamics	Add more temporal vital sign features; interaction terms
Sparse positive class	250 positives in training — SMOTE in this sparse space adds noise	Expand to MIMIC-IV; collect more positive examples

11 Discussion

11.1 Clinical Interpretation

The Random Forest model achieves a recall of 50% — correctly flagging approximately 1 in 2 patients who are about to require mechanical ventilation, 6–12 hours before the event. The AUROC of 0.645 indicates meaningful discrimination above random chance (0.5). For comparison, NEWS2 achieves AUROC values of 0.70–0.80 on similar ICU tasks but requires nurse-administered scoring at fixed intervals, while our model operates continuously on automatically extracted EHR data.

11.2 Comparison with Traditional Approaches

Characteristic	Traditional EWS (NEWS2)	This Project
Update Frequency	Manual — fixed intervals	Continuous — every hour automatically
Data Requirement	Nurse-administered scoring	Automatic extraction from EHR data streams
Temporal Trends	Not captured	3-hour rolling window features (slope, mean, std)
Prediction Horizon	Reactive alert at event time	Predictive — 6–12 hour advance warning
Scalability	Requires nursing time per patient	Fully automated pipeline
Current AUROC	0.70–0.80	0.645 — approaching NEWS2 with 3h features only

11.3 Limitations

- Small cohort: Only 2,520 patients and 69 ventilation events limit statistical power for learning subtle deterioration patterns.
- Short feature window: 3-hour rolling window may not capture deterioration trajectories that develop over longer periods.
- Missing high-value features: Laboratory values (lactate, creatinine, arterial blood gas pH) were unavailable — these are among the strongest predictors in published ICU ML research.
- Precision ceiling at ~1%: All models are limited by feature quality, not model architecture — a fundamental data challenge.
- No external validation: Model was not tested on data from a different hospital or time period — generalizability is unknown.

12 Project Timeline & Weekly Plan

Week	Phase	Key Deliverables & Activities
Week 7	Start & Planning	Finalize project idea · Choose dataset · Define target variable · Set up GitHub repository · Assign team roles · Write project proposal
Week 8	Data Understanding & Cleaning	Load all tables · Create data dictionary · Clean missing values & outliers · Remove duplicates · Fix data types · Document all preprocessing decisions
Week 9	EDA & Baseline Model	Full exploratory data analysis · Create key visualizations · Engineer hourly grid features · Build baseline Logistic Regression · Record baseline results
Week 10	Model Improvement & Comparison	Train Random Forest and XGBoost · Apply all imbalance strategies · Compare 9 model configurations · Document rationale for every change
Week 11	Final Approach & Storytelling	Select final model · Finalize evaluation · Prepare result tables · Choose best visualizations · Improve README · Draft slides
Week 12	Final Polish & Submission	Clean notebooks (top-to-bottom execution) · Finalize README and slides · Full repository review · Rehearse presentation · Submit GitHub link

Project Milestones

Milestone	Target Week	Deliverable
Project Kickoff	Week 7	Proposal document + GitHub repository initialized
Data Pipeline Complete	Week 8	Clean dataset + data dictionary + EDA notebook
Baseline Results	Week 9	Logistic Regression trained + AUPRC/AUROC recorded
Model Comparison Complete	Week 10	All 9 configurations trained and compared
Final Model Selected	Week 11	Best model confirmed + slides draft ready
Final Submission	Week 12	Complete GitHub repo + slides + README submitted

13 Team Roles & Responsibilities

The project is delivered by five members under the supervision of Mentor Courage Dike, with clearly defined primary and additional responsibilities.

Role	Member	Responsibilities
Project Coordinator	Ahmad Assi	Oversee timeline, task distribution, team communication, milestone tracking, and mentor coordination
Data Lead	Ahmad Assi	Load and inspect dataset, handle missing values and duplicates, prepare data dictionary, manage raw data pipeline
EDA & Visualization Lead	Bisan Ghoul	Create all charts and visualizations, explore dataset patterns, write clear analytical insights from EDA
Modeling Lead	Roa'a Jaber & Mohammad Zyoud	Build baseline and improved models, evaluate results, compare all model-strategy configurations
Documentation & Slides Lead	Roaa Abu Arra	Maintain README, organize all project documentation, prepare presentation slide deck
Streamlit Demo Lead	Roa’a Jaber	Develop and deploy interactive Streamlit web application for real-time ICU risk prediction demo
Power BI Dashboard Lead	Roaa Abu Arra	Design and build Power BI dashboard for visual reporting and clinical insights presentation
Deep Learning / LLM Lead	Mohammad Zyoud, Ahmad Assi & Roa'a Jaber	Research, implement, and evaluate advanced deep learning or LLM-based model extensions

Shared Responsibilities — All Members

- Attend all weekly meetings and contribute consistently to project progress
- Participate in reviewing all major technical decisions before implementation
- Contribute to final presentation preparation, rehearsal, and delivery
- Review the complete GitHub repository before final submission

14 Stakeholders

The ICU Deterioration Prediction system has relevance across a broad range of stakeholders in the healthcare sector. Each group stands to benefit differently from the model's deployment:

Stakeholder	Interest	Primary Benefit
Hospitals	Reduce ICU mortality and optimize resource use	Deploy early warning system — flag at-risk patients 6–12h in advance
Clinics	Triage high-risk patients before ICU transfer	Use risk scores to prioritize escalation and transfer decisions
Physicians & Nurses	Real-time decision support; reduced cognitive overload	Data-driven risk scores to supplement clinical judgment
Medical Analysts & Researchers	Population-level analysis of deterioration patterns	Feature importance and SHAP values for clinical research insights
Healthcare Decision-Makers	ICU staffing optimization and system-level policy	Aggregated risk data to guide resource allocation
Data Scientists	Benchmark for ICU prediction methodology	Open, reproducible pipeline on publicly available eICU dataset

> The system is designed to augment — not replace — clinical judgment. It provides a continuous, automated second opinion that surfaces patterns invisible to manual monitoring.

15 Future Work & Recommendations

15.1 Immediate Improvements (High Priority)

#	Area	Recommended Action	Expected Impact
1	Feature Windows	Extend rolling windows from 3h to 6h and 12h	Biggest expected performance gain
2	Vital Trends	Add SpO2 and HR deterioration slope over 6h and 12h	Improve temporal signal quality
3	Interaction Features	Add combined signals: low SpO2 + high RR, rising FiO2 + rising HR	Capture multi-parameter clinical patterns
4	Label Verification	Confirm y_high=1 rows are placed correctly relative to vital sign timestamps	Ensure label accuracy

15.2 Medium-Term Extensions

- SHAP Interpretability: Individual patient explanations showing clinicians exactly which features drove each risk prediction.
- Streamlit Demo Application: Interactive web application where clinicians select a patient, view hourly risk trajectory, and see contributing features.
- Power BI Dashboard: Executive-level reporting showing ICU-wide risk distribution, trend over time, and model performance metrics.
- Prospective Validation: Testing the model on new, unseen ICU data from a different institution to assess generalizability.

15.3 Long-Term Research

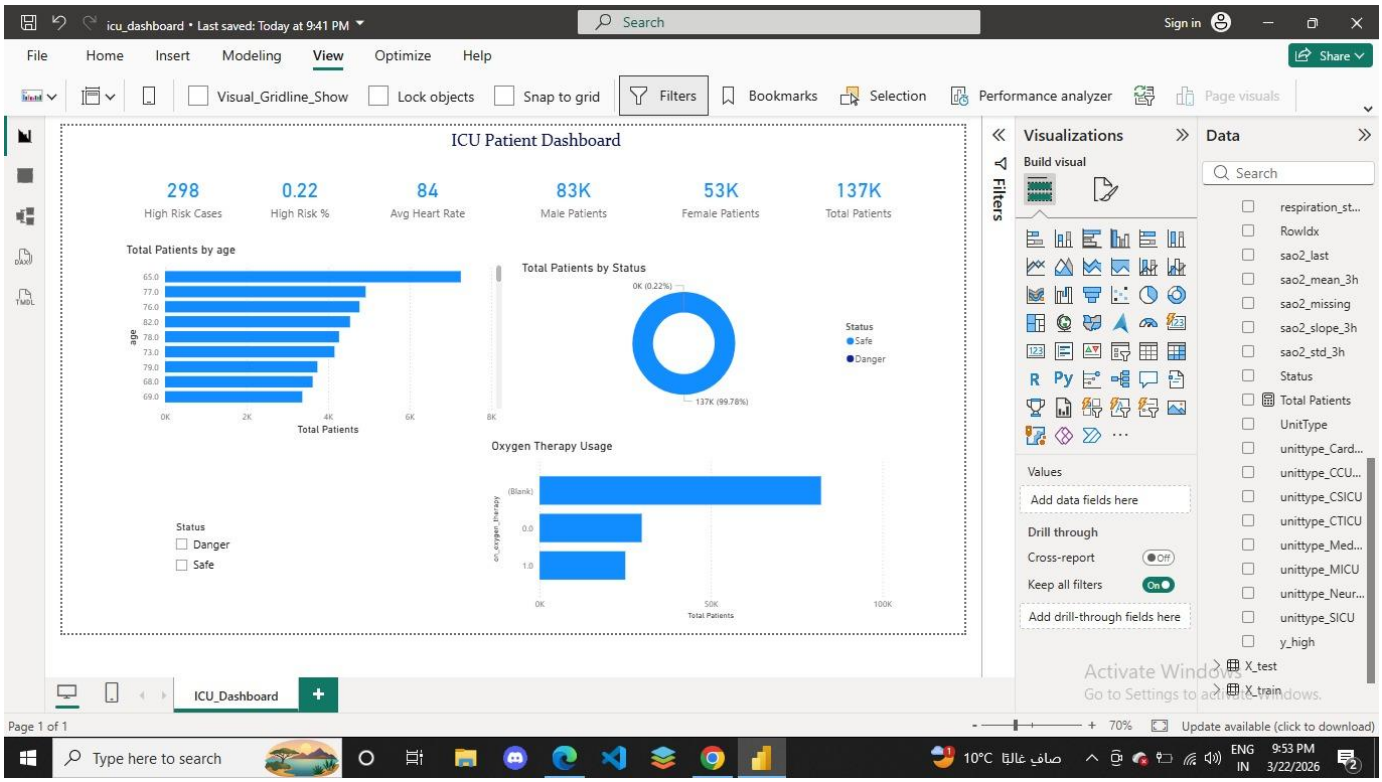
- MIMIC-IV Scaling: Scaling the pipeline to MIMIC-IV (100,000+ ICU stays) to enable deep learning approaches and robust statistical conclusions.
- LSTM / Temporal Neural Networks: Once the dataset is large enough, LSTM networks can learn temporal patterns directly from raw time series.
- LLM Integration: Incorporating clinical notes via large language model embeddings to add qualitative context beyond vital sign numerics.

15.4 Power BI Dashboard

As part of the project's clinical reporting and stakeholder communication deliverables, an interactive Power BI dashboard was built on top of the engineered dataset (X_train and X_test combined). The dashboard loads the 136,867-row hourly patient grid and the engineered features — including vital sign rolling statistics, respiratory parameters, treatment flags, and the binary risk label (y_high) — to provide a visual analytics layer over the machine learning pipeline outputs. The dashboard enables clinical and operational stakeholders to explore patient risk distribution, demographic breakdowns, and oxygen therapy patterns without requiring direct interaction with the underlying model code.

Dashboard View 1 — Full Dataset Overview

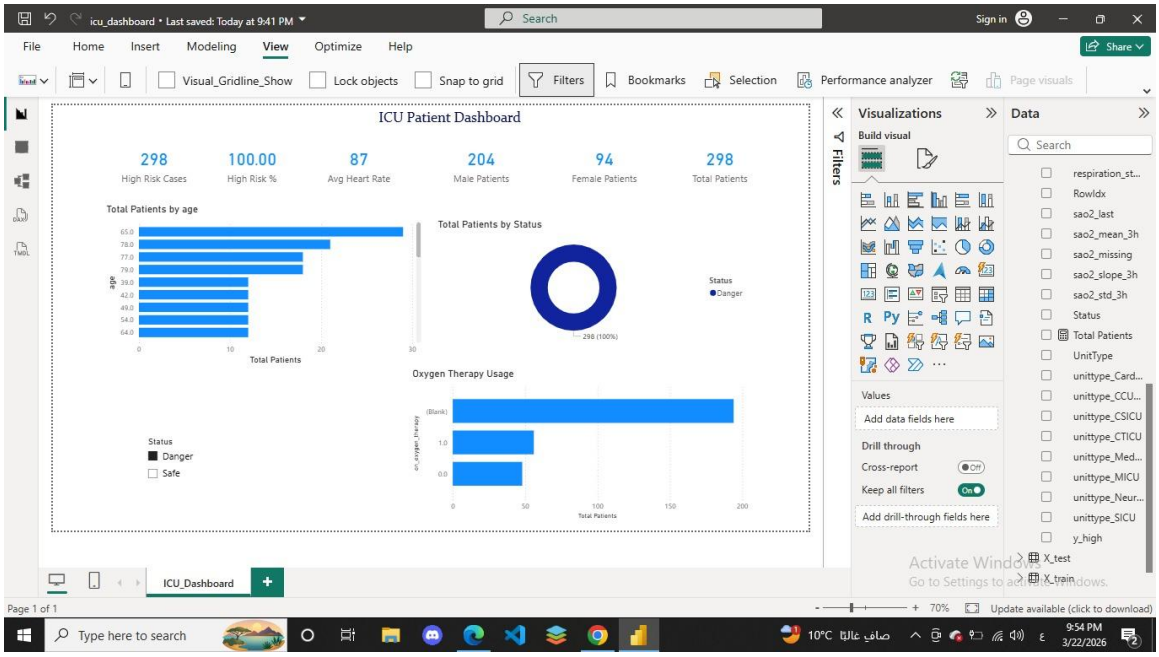
The first view presents the unfiltered dataset. The KPI cards at the top surface the key model-level statistics: 298 high-risk patient-hours (0.22% of the total), an average heart rate of 84 bpm across all windows, 83K male and 53K female patient windows, and 137K total patient records. The "Total Patients by Age" horizontal bar chart shows that the highest-volume age group is 65 years, reflecting the typical ICU population demographics confirmed in the EDA. The "Total Patients by Status" donut chart illustrates the extreme class imbalance visually: 99.78% of windows are labelled Safe (blue) and only 0.22% are labelled Danger (darker blue) — the core challenge that drove the use of AUPRC as the primary evaluation metric. The "Oxygen Therapy Usage" bar chart breaks down patient volumes across therapy values (0.0 = no therapy, 1.0 = on therapy, Blank = not recorded), showing that the majority of windows have no active oxygen therapy flag — consistent with the 74% coverage figure found during feature engineering.



Dashboard View 2 — Filtered: Safe Windows Only

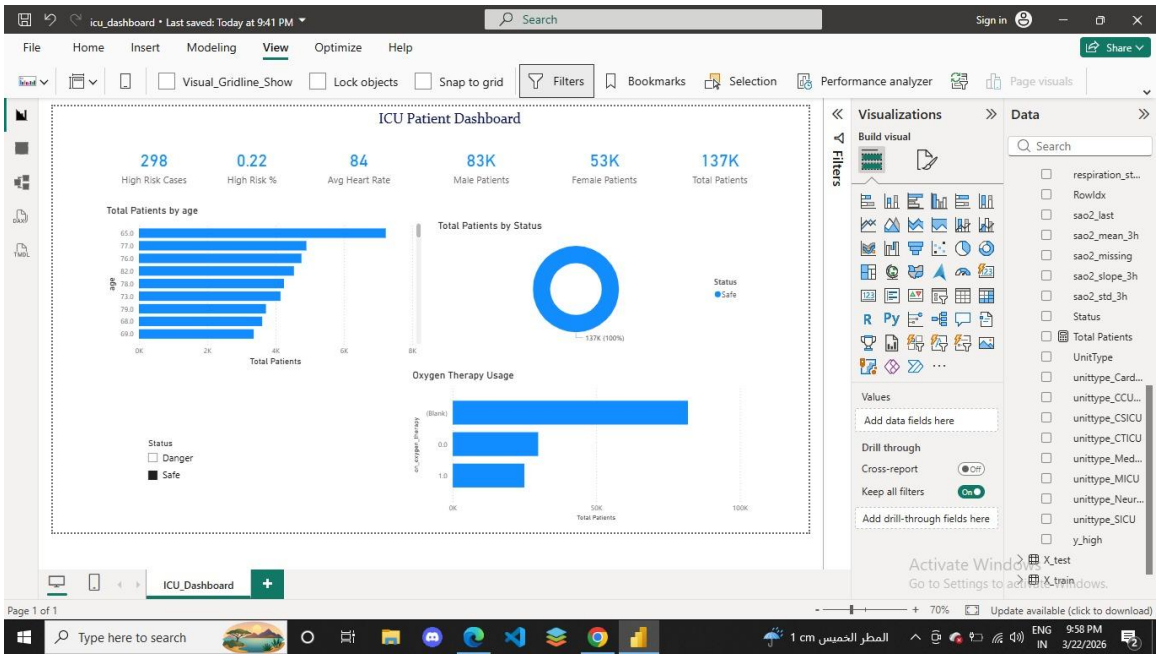
The second view applies a filter to Status = Safe, isolating the 136,569 windows where no ventilation was required in the 6–12 hour window. The donut chart now shows 100% Safe (137K, 100.00%), confirming the filter is working correctly and the Safe class contains no contamination from labelled Danger windows. The KPI cards remain consistent (298 high-risk cases, 84 avg heart rate, 83K/53K male/female split) since these aggregate the

full dataset metadata. The age distribution is visually similar to the unfiltered view, as expected: the Safe class comprises 99.78% of all records and its demographic shape dominates the overall distribution. The oxygen therapy bar chart preserves the same breakdown as View 1, reinforcing that oxygen therapy usage is broadly distributed across the dataset rather than being concentrated in the minority class.



Dashboard View 3 — Filtered: Danger Windows Only

The third view filters to Status = Danger only, isolating the 298 true high-risk windows across 69 ventilated patients. Several clinically meaningful differences emerge compared to View 1. The total patient count drops to 298 and the High Risk % reads 100.00%, confirming the filter captures only the labelled Danger windows. The average heart rate rises to 87 bpm (vs 84 bpm overall), consistent with the bivariate analysis finding that tachycardia is a marker of approaching ventilation. The age distribution shifts: the highest-volume age groups in the danger cohort are 65 and 78 years, with younger ages (39, 42, 49) also prominent — suggesting that for this subgroup, younger patients who do deteriorate tend to reach the danger threshold earlier in their stay. The donut chart now shows 298 windows at 100% Danger (all dark). The oxygen therapy chart shows a reversal in the blank/1.0 ratio compared to the Safe view: a higher proportion of Danger windows have on_oxygen_therapy = 1.0, directly supporting the finding that oxygen therapy is a significant escalation signal in the pre-ventilation period and confirming the clinical validity of including it as a binary feature.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Male Patients
53K
Female Patients
137K
Total Patients
Total Patients by age
Total Patients by Status
Oxygen Therapy Usage
Status
Danger
Safe
Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Keep all filters
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sao2_mean_3h
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16 Innovation & Differentiation

This project differentiates itself from static clinical scoring systems through methodological rigor, continuous monitoring architecture, and strict leakage prevention.

Dimension	Traditional Systems (NEWS2)	This Project
Update Frequency	Manual — fixed intervals	Continuous — automated every hour
Temporal Modeling	Static snapshot	Dynamic rolling window — trends, slopes, variability
Prediction Horizon	Reactive at event time	Predictive — 6–12 hour advance warning
Feature Complexity	Fixed 6–7 vital parameters	31 engineered features
Data Pipeline	Manual nursing observation	Automated EHR extraction and alignment
Scalability	Requires nursing time per patient	Fully automated — scales to entire ICU
Leakage Prevention	N/A (rule-based)	Patient-level split + merge_asof + vent_start_min exclusion
AUROC (this dataset)	~0.70–0.80 (with nurse assessment)	0.645 (fully automated, 3h features only)

17 Conclusion

This project successfully demonstrated that machine learning models trained on the eICU Collaborative Research Database can predict ICU patient deterioration — specifically the need for mechanical ventilation — up to 6–12 hours in advance with meaningful discrimination above random chance.

The best-performing configuration — Random Forest with balanced class weights — achieved an AUROC of 0.645 and a recall of 50%, correctly flagging half of all true high-risk windows. This is a clinically significant result given the extreme class imbalance (436:1) and the limited 3-hour feature window used.

The complete end-to-end pipeline — from raw eICU CSV files through feature engineering, model training, and evaluation — was built to professional research standards: strict leakage prevention at every stage, patient-level train/test split, and AUPRC as the primary metric appropriate for severe class imbalance.

> **The methodology developed in this project — time-aware hourly evaluation, strict leakage prevention via merge_asof, patient-level splitting, and AUPRC-focused evaluation — represents a complete, reproducible, and clinically motivated framework for ICU deterioration prediction.**

Acknowledgements

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The following primary sources informed the design, methodology, and clinical context of this project:

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